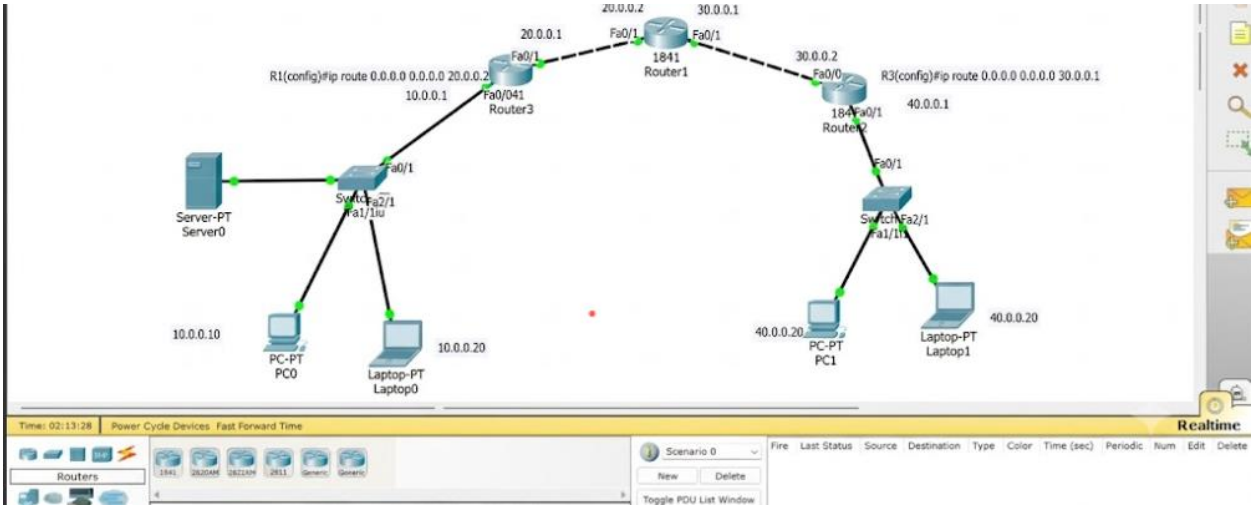
USN:1WN24CS186

4.How to Configure Static Route and Default Route to the Routers.

Solution:




```
R1>enable
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 0.0.0.0 0.0.0.0 20.0.0.2
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
R1#
```

```
R2>enable
R2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.2
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#write memory
Building configuration...
[OK]
R2#
```

Copy Paste

```
R3>enable
R3#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip route 0.0.0.0 0.0.0.0 30.0.0.1
R3(config)#exit
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#write memory
Building configuration...
[OK]
R3#
```

```
R1>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 20.0.0.2 to network 0.0.0.0

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    20.0.0.0/8 is directly connected, FastEthernet0/1
S*   0.0.0.0/0 [1/0] via 20.0.0.2
R1>
```

Copy Paste

```
R2>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
C    20.0.0.0/8 is directly connected, FastEthernet0/0
C    30.0.0.0/8 is directly connected, FastEthernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2
R2>
```

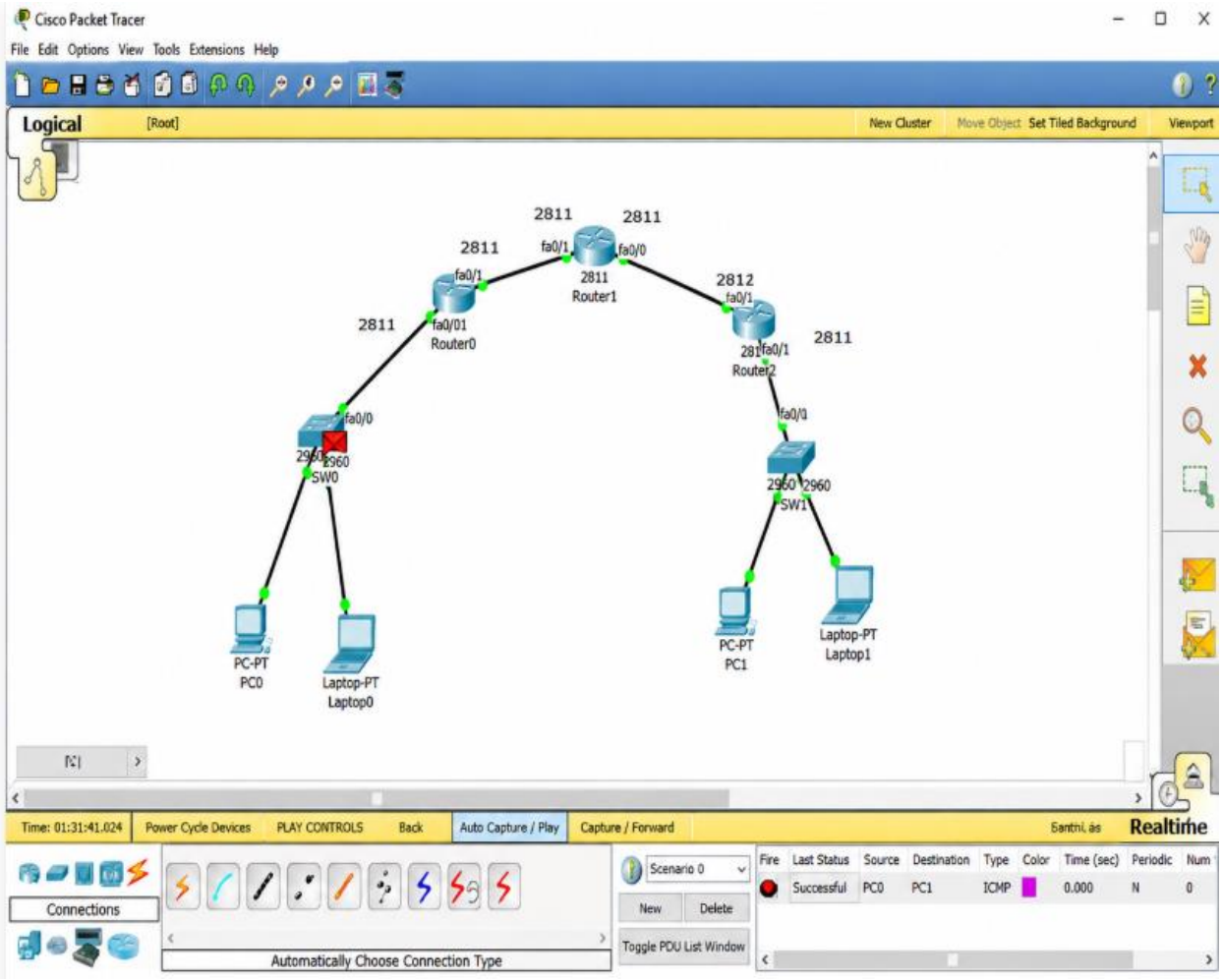
```
Packet Tracer PC Command Line 1.0
PC>ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 40.0.0.10: bytes=32 time=151ms TTL=125
Reply from 40.0.0.10: bytes=32 time=157ms TTL=125
Reply from 40.0.0.10: bytes=32 time=172ms TTL=125
Reply from 40.0.0.10: bytes=32 time=190ms TTL=125

Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 151ms, Maximum = 190ms, Average = 167ms

PC>
```


Cisco Packet Tracer
File Edit Options View Tools Extensions Help
Logical
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
2811
2811
2811
fa0/1
fa0/0
2811
Router1
2812
fa0/1
2811
fa0/1
fa0/01
Router0
2811
fa0/1
Router2
fa0/0
fa0/3
2960
SW0
2960
2960
SW1
PC-PT
PC0
Laptop-PT
Laptop0
PC-PT
PC1
Laptop-PT
Laptop1
Time: 01:31:41.024
Power Cycle Devices
PLAY CONTROLS
Back
Auto Capture / Play
Capture / Forward
Realtime
Connections
Automatically Choose Connection Type
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time (sec)
Periodic
Num
Successful
PC0
PC1
ICMP
0.000
N
0

5/10/20

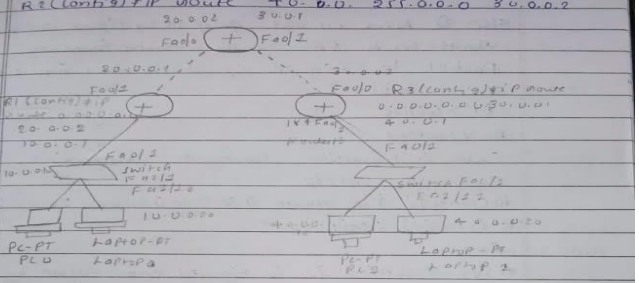
Date Page

4) How to Configure static Route & Duplicate Route to the Routers

Step 1) : Create the topology with 2 subnetworks apply the basic configuration to all both of interfaces

R2(config) # ip route 10.0.0.0 255.0.0.0 20.0.0.1

R2(config) # ip route 40.0.0.0 255.0.0.0 30.0.0.2



Step 2): Configure static on R2 to route packets for 10.0.0.0 and 40.0.0.0 networks using the commands in CLI

R2(config) # ip route Destination / network

R2>enable

R2 # config t

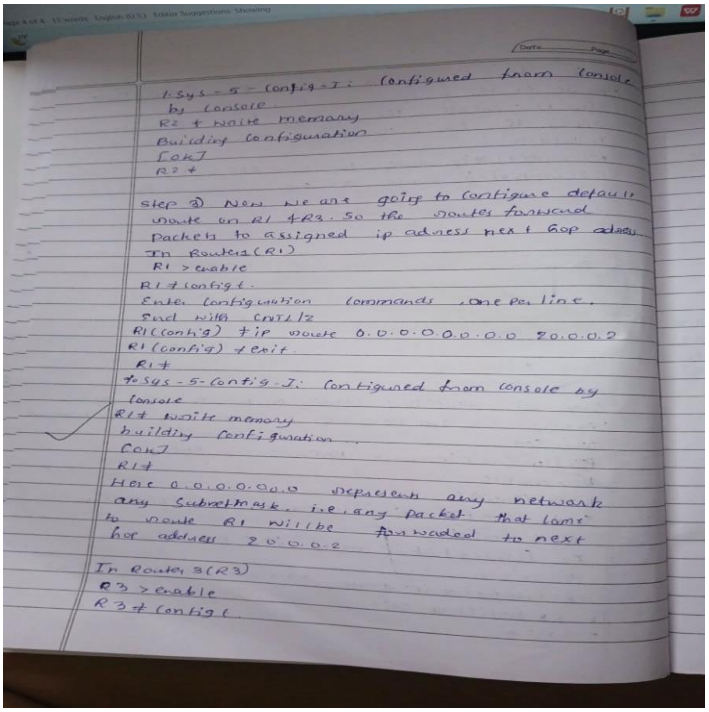
Enter configuration commands, one per line. End with CNTL/Z

R2 [config] # ip route 10.0.0.0 255.0.0.0 20.0.0.1

R2 [config] # ip route 40.0.0.0 255.0.0.0 30.0.0.2

R2 (config) # exit.

R2 #



Date Page

1 Sys - 5 - config - I : Configured from console by console

R2 # write memory

Building configuration

[OK]

R2 #

Step 3) Now we are going to configure default route on R1 & R3. So the routes forwarded packets to assigned ip address next hop address

In Router1 (R1)

R1 > enable

R1 # config t.

Enter configuration commands, one per line.

End with CNTL/Z

R1(config) # ip route 0.0.0.0 0.0.0.0 20.0.0.2

R1 (config) # exit.

R1 #

%Sys - 5 - config - I: Configured from console by console

R1 # write memory

building configuration...

[OK]

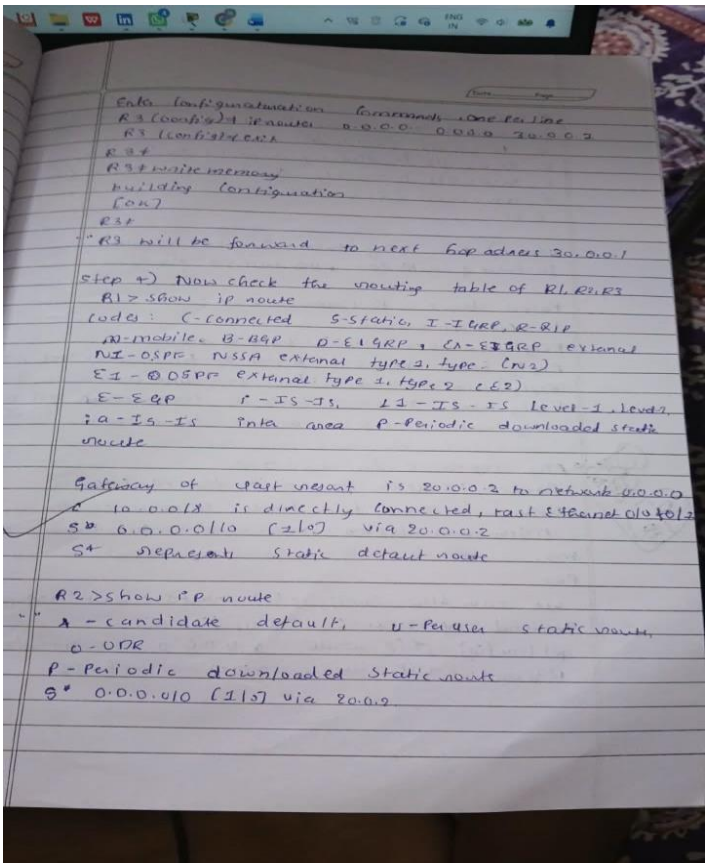
R1 #

Here 0.0.0.0 0.0.0.0 represent any network any subnetmask. i.e, any packet that came to route R1 will be forwarded to next hop address 20.0.0.2

In Router3 (R3)

R3 > enable

R3 # config t



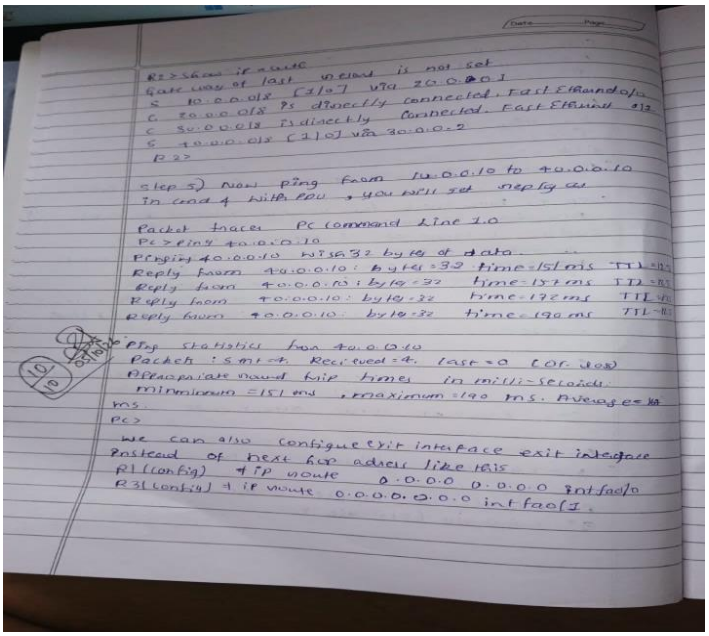
Enter configuration commands, one per line
R3 (config)# ip route 0.0.0.0 0.0.0.0 20.0.0.2
R3 (config)# exit
R3#
R3# write memory
Building configuration
[OK]
R3#

"R3 will be forward to next hop adress 30.0.0.1

Step 4) Now check the routing table of R1, R2, R3
R1> show ip route
Codes: C-connected S-static, I-IGRP, R-RIP
M-mobile, B-BGP D-EIGRP, EX-EIGRP external
O-OSPF NSSA external type 1, type 2 (N2)
E1-OSPF external type 1, type 2 (E2)
E-EGP i-IS-IS, L1-IS-IS level-1, level-2,
ia-IS-IS inta area *-Periodic downloaded static
route

Gateway of last resort is 20.0.0.2 to network 0.0.0.0
C 10.0.0.0/8 is directly connected, FastEthernet 0/0
S* 0.0.0.0/0 [1/0] via 20.0.0.2
S* represents static default route

R2>show ip route
" * - candidate default, U-Per user static route,
O-ODR
P-Periodic downloaded static route
S* 0.0.0.0/0 [1/5] via 20.0.0.2



R2>show ip route
Gateway of last resort is not set
S 10.0.0.0/8 [1/0] via 20.0.0.1
C 20.0.0.0/8 is directly connected, FastEthernet 0/0
C 30.0.0.0/8 is directly connected, FastEthernet 0/1
S 40.0.0.0/8 [1/0] via 30.0.0.2
R2>

Step 5) Now ping from 10.0.0.10 to 40.0.0.10
In cmd of WithPDU → you will get reply as

Packet tracer PC command Line 1.0
PC>ping 40.0.0.10
Pinging 40.0.0.10 with 32 bytes of data:
Reply from 40.0.0.10: bytes=32 time=151ms TTL=125
Reply from 40.0.0.10: bytes=32 time=151ms TTL=125
Reply from 40.0.0.10: bytes=32 time=172ms TTL=125
Reply from 40.0.0.10: bytes=32 time=190ms TTL=125

Ping statistics for 40.0.0.10
Packets: Sent=4, Received=4, Lost=0 (0% loss)
Approximate round trip times in milli-seconds:
Minimum=151 ms, Maximum=190 ms, Average=166
ms
PC>

We can also configure exit interface exit interface
Instead of next hop adress like this
R1(config) # ip route 0.0.0.0 0.0.0.0 int fa0/0
R3(config) # ip route 0.0.0.0 0.0.0.0 int fa0/1